# Official proposed theorem

Conditional regularity criterion with filled classical lemmas
and named obstructions on the remaining slots

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This document does not claim the Clay Millennium statement.

**Abstract**

The official proposed theorem of this redraft is a *conditional* regularity criterion on the exact three-dimensional incompressible Navier–Stokes equations. The Clay Mathematics Institute problem on Navier–Stokes existence and smoothness *is* the problem of global regularity for that system. They are one statement, not two enigmas. Classical lemmas (transport cancellation, viscous dissipation, algebraic stretching identity, Young at $\kappa = 3/2$, Hölder, Sobolev embedding on $\mathbb{T}^3$, and the published Kato and Beale–Kato–Majda theorems) are proved or cited with complete proofs. The lemmas that would convert the criterion into the Clay statement—a genuine strain endpoint, stretching identification with the angular kernel and vanishing commutators, a majorant whose coefficients depend only on initial data, and the mollifier limit—are expanded to the obstruction and then stopped. Chat fragments that contradict the exact PDE or reintroduce $\kappa = 1$ are not used.

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# 1 Clay statement and global regularity are the same problem

Fefferman's official formulation asks whether, for every smooth divergence-free initial velocity of finite energy, the three-dimensional incompressible Navier–Stokes equations admit a unique solution that remains smooth for all positive times. That is precisely global regularity of the Cauchy problem. There is no separate "enigmatic process of incompressible flow" whose resolution would produce global regularity as an accidental by-product, and no separate global-regularity enigma whose resolution would produce Navier–Stokes smoothness as a by-product. A proof of one is a proof of the other. A criterion that assumes a bound in order to conclude a bound is neither.

The official proposed theorem below is therefore not a second problem. It is the Clay statement with its unfinished hypotheses written as named lemmas. Filling those hypotheses with first-principles, non-circular, solution-independent estimates *would* be the Clay proof. Declaring them filled because the names coincide is circular.

# 2 Official proposed theorem

**Theorem 2.1** (Proposed theorem of the redraft — conditional criterion)**.** *Let $s > 5/2$ and let $u_0 \in H^s(\mathbb{T}^3;\mathbb{R}^3)$ satisfy $\operatorname{div} u_0 = 0$. Let $u$ be the unique Kato solution of incompressible Navier–Stokes*

$$\partial_t u + (u\cdot\nabla)u + \nabla p = \nu\Delta u, \qquad \operatorname{div} u = 0, \qquad u(\,\cdot\,,0) = u_0, \tag{1}$$

*on its maximal classical interval $[0,T_*)$, with vorticity $\omega = \operatorname{curl} u$ obeying*

$$\partial_t\omega + (u\cdot\nabla)\omega = (\omega\cdot\nabla)u + \nu\Delta\omega. \tag{2}$$

*Assume Lemmas 4.1, 4.2, 4.3, 4.4, 4.5, and 4.6 with constants depending only on $\|u_0\|_{H^s}$ and $\nu$. Then*

$$\int_0^{T_*} \|\omega(\,\cdot\,,t)\|_{L^\infty}\,dt < \infty,$$

*hence $T_* = \infty$ by Lemma 3.9, and $u$ remains smooth for all $t \geq 0$.*

**Remark 2.2** (Status)**.** Theorem 2.1 is the official proposed theorem of this redraft. It is not the Clay theorem until every cited lemma is a theorem. The lemmas marked "classical, filled" below are theorems. The lemmas marked "obstruction" are not.

# 3 Filled classical lemmas

Throughout this section, integrals without a domain are over $\mathbb{T}^3 = \mathbb{R}^3/(2\pi\mathbb{Z})^3$, and boundary terms vanish by periodicity.

## 3.1 Algebraic structure of stretching

**Lemma 3.1** (Stretching is the strain quadratic form)**.** *Let $u \in C^1(\mathbb{T}^3;\mathbb{R}^3)$ and $\omega = \operatorname{curl} u$. Write $S = \frac{1}{2}(\nabla u + (\nabla u)^\top)$ and $\Omega = \frac{1}{2}(\nabla u - (\nabla u)^\top)$. Then $\Omega v = \frac{1}{2}\omega\times v$ for every vector $v$, and*

$$\langle\omega, (\omega\cdot\nabla)u\rangle = \langle\omega, S\omega\rangle.$$

*Proof.* The decomposition $\nabla u = S + \Omega$ is the unique symmetric/skew splitting. For the skew part, the identity

$$(\Omega v)_i = \frac{1}{2}\sum_j (\partial_j u_i - \partial_i u_j)v_j = \frac{1}{2}(\omega\times v)_i$$

holds by the definition $\omega_k = \varepsilon_{k\ell m}\partial_\ell u_m$ and the contraction $\varepsilon_{ijk}\varepsilon_{k\ell m} = \delta_{i\ell}\delta_{jm} - \delta_{im}\delta_{j\ell}$. Hence

$$(\omega \cdot \nabla)u = S\omega + \Omega\omega = S\omega + \tfrac{1}{2}\omega \times \omega = S\omega,$$

and taking the inner product with $\omega$ yields the claim because $\langle \omega, \omega \times \omega \rangle = 0$. □

### 3.2 Transport cancellation

**Lemma 3.2** (Convective cancellation on $\mathbb{T}^3$)**.** *Let $u \in C^1(\mathbb{T}^3;\mathbb{R}^3)$ with $\operatorname{div} u = 0$ and let $\omega \in C^1(\mathbb{T}^3;\mathbb{R}^3)$. Then*

$$\int_{\mathbb{T}^3} \langle \omega, (u \cdot \nabla)\omega \rangle \, d\lambda = 0. \tag{3}$$

*If $\varphi \in C^1(\mathbb{T}^3)$ and $\partial_t\varphi + u \cdot \nabla\varphi = 0$, the same cancellation holds for the weighted densities $\frac{1}{2}|\omega|^2\varphi$ and $\frac{1}{4}|\omega|^4\varphi$ up to the explicit product-rule terms recorded in the proof.*

*Proof.* Pointwise,

$$\langle \omega, (u \cdot \nabla)\omega \rangle = \frac{1}{2} u \cdot \nabla(|\omega|^2) = \frac{1}{2} \operatorname{div}(u|\omega|^2) - \frac{1}{2}(\operatorname{div} u)|\omega|^2.$$

The second summand vanishes by $\operatorname{div} u = 0$. Integrating the first over $\mathbb{T}^3$ and applying the divergence theorem on the torus gives

$$\int_{\mathbb{T}^3} \operatorname{div}(u|\omega|^2) \, d\lambda = 0.$$

This is (3).

For a transported scalar $\varphi$ with $\partial_t\varphi + u \cdot \nabla\varphi = 0$,

$$\partial_t(\tfrac{1}{2}|\omega|^2\varphi) + u \cdot \nabla(\tfrac{1}{2}|\omega|^2\varphi) = \varphi\langle \omega, \partial_t\omega + (u \cdot \nabla)\omega \rangle.$$

The left side is a divergence after using $\operatorname{div} u = 0$, hence integrates to zero. The same computation with $|\omega|^4$ uses

$$\partial_t|\omega|^4 = 4|\omega|^2\langle \omega, \partial_t\omega \rangle, \tag{4}$$

which is the chain rule. Thus the transport contribution to $\frac{d}{dt}\int \frac{\kappa}{4}|\omega|^4\varphi$ reduces to

$$\kappa \int |\omega|^2\varphi \, \langle \omega, \partial_t\omega + (u \cdot \nabla)\omega \rangle \, d\lambda$$

and does not itself produce a stretching or dissipation term. □

### 3.3 Viscous dissipation

**Lemma 3.3** (Viscous integration by parts)**.** *Let $\omega \in C^2(\mathbb{T}^3;\mathbb{R}^3)$. Then*

$$\int_{\mathbb{T}^3} \langle \omega, \Delta\omega \rangle \, d\lambda = -\int_{\mathbb{T}^3} |\nabla\omega|^2 \, d\lambda \le 0. \tag{5}$$

*Proof.* On each Cartesian component $\omega_i$,

$$\int_{\mathbb{T}^3} \omega_i \Delta\omega_i \, d\lambda = \int_{\mathbb{T}^3} \omega_i \sum_{j=1}^{3} \partial_{jj}\omega_i \, d\lambda = -\int_{\mathbb{T}^3} \sum_{j=1}^{3} (\partial_j\omega_i)^2 \, d\lambda,$$

where the boundary terms vanish by periodicity. Summing on $i$ yields (5). □

### 3.4 Coefficient algebra at $\kappa = 3/2$

**Lemma 3.4** (Official scale identities)**.** *Define*

$$C_{\mathrm{CZ}}(3) := \frac{3}{8\pi} \cdot 4\pi = \frac{3}{2}, \qquad \kappa := C_{\mathrm{CZ}}(3) = \frac{3}{2}.$$

*Then*

$$4\kappa = 6, \qquad \frac{\kappa}{4} = \frac{3}{8}, \qquad \kappa' := \frac{3}{4}\kappa = \frac{9}{8}.$$

*These identities replace every former stand-in $\kappa = 1$. They do not assert that the operator norm of $\nabla \circ K_{\mathrm{BS}}$ on $L^\infty$ equals $3/2$.*

*Proof.* The product $\frac{3}{8\pi} \cdot 4\pi = \frac{3}{2}$ is cancellation of $\pi$. The three companions are arithmetic. The last sentence is the distinction between a spherical prefactor and a Calderón–Zygmund endpoint; see Lemma 4.1. □

### 3.5 Hölder and Young

**Lemma 3.5** (Hölder gate on $\mathbb{T}^3$)**.** *Let $\varphi \in L^\infty(\mathbb{T}^3)$ and $\omega \in L^6(\mathbb{T}^3)$. Then*

$$\int_{\mathbb{T}^3} |\omega|^4 |\varphi| \, d\lambda \;\le\; \|\varphi\|_{L^\infty} \Big( \int_{\mathbb{T}^3} |\omega|^6 \, d\lambda \Big)^{2/3} \lambda(\mathbb{T}^3)^{1/3}. \tag{6}$$

*Proof.* Write $|\omega|^4|\varphi| = |\varphi| \cdot |\omega|^4 \cdot 1$. Hölder with conjugate exponents $3/2$ and $3$ on the pair $(|\omega|^4, 1)$ gives

$$\int |\omega|^4 \;\le\; \Big( \int |\omega|^6 \Big)^{2/3} \Big( \int 1^3 \Big)^{1/3} = \Big( \int |\omega|^6 \Big)^{2/3} \lambda(\mathbb{T}^3)^{1/3}.$$

Multiplying by $\|\varphi\|_{L^\infty}$ yields (6). □

**Lemma 3.6** (Young at the official parameter)**.** *Let $a, b \ge 0$ and let $\varepsilon = 3/8$. Then*

$$ab \;\le\; \frac{3}{8} a^{3/2} + C_{3/8}\, b^3, \qquad C_{3/8} = \frac{2}{3} \Big( \frac{8}{3} \Big)^{1/2} = \frac{2}{3} \cdot \frac{2\sqrt{2}}{\sqrt{3}} = \frac{4\sqrt{2}}{3\sqrt{3}}. \tag{7}$$

*Consequently, writing $I_4(\varphi) = \int |\omega|^4 |\varphi| \, d\lambda$ and $I_6 = \int |\omega|^6 \, d\lambda$,*

$$6M I_4(\varphi) \;\le\; \frac{3}{8} I_6 + C_{\mathrm{Y}} M^3 \|\varphi\|_{L^\infty}^{3/2} \lambda(\mathbb{T}^3)^{1/2}, \tag{8}$$

*for an explicit $C_{\mathrm{Y}}$ assembled from* (6) *and* (7)*,* provided *one already has a pointwise or integrated stretching bound of size $6MI_4(\varphi)$. That proviso is Lemma 4.2, not this lemma.*

*Proof.* Young's inequality for conjugate exponents $p = 3/2$ and $q = 3$ reads

$$ab \le \varepsilon \frac{a^p}{p} + C_\varepsilon \frac{b^q}{q} \quad \text{after rescaling,}$$

or, in the elementary form obtained from $xy \le \frac{x^p}{p} + \frac{y^q}{q}$ with $x = \varepsilon^{1/p} a$ and $y = \varepsilon^{-1/p} b$,

$$ab \le \varepsilon \frac{a^{3/2}}{3/2} + \frac{2}{3} \varepsilon^{-1/2} b^3 = \frac{2}{3} \varepsilon a^{3/2} + \frac{2}{3} \varepsilon^{-1/2} b^3.$$

The tree's official parameter is the scale tag $\varepsilon_{\mathrm{Y}} = \kappa/4 = 3/8$ applied to the integrated stretching form, not necessarily this abstract $p = 3/2$ slot. The identity (7) records one consistent elementary Young pair. Substituting $a = I_6^{2/3}$ and $b = 6M\|\varphi\|_{L^\infty} \lambda(\mathbb{T}^3)^{1/3}$ into (6) and applying a Young pair produces a bound of the shape (8). The numerical factor $C_{\mathrm{Y}}$ is elementary once the pair is fixed. What this lemma does *not* supply is the justification that stretching contributes a term of size $6MI_4(\varphi)$ rather than a larger multiple involving the true endpoint constant of $\nabla \circ K_{\mathrm{BS}}$. □

**Lemma 3.7** (Sobolev embedding on $\mathbb{T}^3$)**.** *There exists $C_{\mathrm{Sob}} = C_{\mathrm{Sob}}(\mathbb{T}^3)$ such that*

$$\|f\|_{L^6(\mathbb{T}^3)} \;\le\; C_{\mathrm{Sob}}\|f\|_{H^1(\mathbb{T}^3)} \qquad \text{for all } f \in H^1(\mathbb{T}^3).$$

*Proof.* This is the Sobolev embedding $H^1(\mathbb{T}^3) \hookrightarrow L^6(\mathbb{T}^3)$, which holds because $1 - 3/2 = -1/2$ and $L^6$ is the critical Lebesgue space for $H^1$ in dimension three (equivalently $W^{1,2}(\mathbb{T}^3) \hookrightarrow L^6(\mathbb{T}^3)$). A complete first-principles proof proceeds from the Gagliardo–Nirenberg–Sobolev inequality on $\mathbb{R}^3$ applied to a periodic extension, or from Fourier series

$$\|f\|_{L^6} \le C\|\langle D\rangle f\|_{L^2}, \qquad \langle D\rangle = (1-\Delta)^{1/2},$$

which is a Mihlin multiplier theorem on $\mathbb{T}^3$. We record it as a classical embedding with a domain-dependent constant and do not treat $C_{\mathrm{Sob}}$ as an open Frohmanian object. □

### 3.6 Published local existence and continuation

**Lemma 3.8** (Kato local existence)**.** *Let $s > 5/2$ and $u_0 \in H^s(\mathbb{T}^3;\mathbb{R}^3)$ with $\operatorname{div} u_0 = 0$. There exist $T_{\mathrm{loc}} = T_{\mathrm{loc}}(\|u_0\|_{H^s}, \nu) > 0$ and a unique solution*

$$u \in C\big([0,T_{\mathrm{loc}}];H^s\big) \cap C^1\big([0,T_{\mathrm{loc}}];H^{s-2}\big)$$

*of* (1)*.*

*Proof.* This is the standard Kato–Fujita local theory in Sobolev spaces above the critical index. Mild solutions of the integral equation

$$u(t) = e^{\nu t\Delta}u_0 - \int_0^t e^{\nu(t-\tau)\Delta}\Pi\operatorname{div}(u\otimes u)(\tau)\,d\tau$$

are constructed by a Picard contraction in

$$C\big([0,T];H^s\big)$$

for $T$ small enough that

$$C\|u_0\|_{H^s}^2 T^{1/2}$$

is less than $\|u_0\|_{H^s}$, using the estimate

$$\|\Pi\operatorname{div}(u\otimes u)\|_{H^{s-2}} \le C\|u\|_{H^s}^2$$

and the analytic bound $\|e^{\nu t\Delta}\|_{H^{s-2}\to H^s} \le C(\nu t)^{-1}$. Uniqueness in this class is likewise standard. The existence time depends on $\|u_0\|_{H^s}$ and $\nu$ only. We cite Fujita–Kato (1964) and Kato (1972) and treat the lemma as classical. A Lean transcription remains a software remainder; it is not a mathematical remainder. □

**Lemma 3.9** (Beale–Kato–Majda continuation)**.** *Let $u$ be a classical solution of* (1) *on a maximal interval $[0,T_*)$. If*

$$\int_0^{T_*}\|\omega(\,\cdot\,,t)\|_{L^\infty}\,dt < \infty,$$

*then $T_* = \infty$.*

*Proof.* The Beale–Kato–Majda criterion states that the $H^s$ norm, $s > 5/2$, remains finite on $[0,T]$ whenever $\int_0^T\|\omega\|_{L^\infty}\,dt$ is finite. The logarithmic Sobolev estimate

$$\|\nabla u\|_{L^\infty} \le C\big(1 + \|\omega\|_{L^\infty}\log(1+\|u\|_{H^s})\big)$$

together with the energy inequality for $\|u\|_{H^s}$ produces

$$\frac{d}{dt}\|u\|_{H^s} \le C\|\omega\|_{L^\infty}\|u\|_{H^s}\log(1+\|u\|_{H^s}),$$

which integrates to a finite bound on any interval where $\int\|\omega\|_{L^\infty}$ is finite. Maximality of $T_*$ therefore forces $T_* = \infty$. We cite Beale–Kato–Majda, *Commun. Math. Phys.* 94 (1984), and treat the lemma as classical. □

### 3.7 Elementary Biot–Savart identities

**Lemma 3.10** (Curl of a gradient vanishes)**.** *For $p \in C^2(\mathbb{T}^3)$,* $\operatorname{curl} \nabla p = 0$.

*Proof.* Mixed partials commute: $(\operatorname{curl} \nabla p)_i = \varepsilon_{ijk}\partial_j\partial_k p = 0$ by $\partial_j\partial_k = \partial_k\partial_j$. □

**Lemma 3.11** (Divergence of Biot–Savart, Fourier form)**.** *On $\mathbb{T}^3$, if $\omega \in C^\infty$ and* $\operatorname{div}\omega = 0$*, and if*

$$\hat{u}(\xi) = \frac{i\xi}{|\xi|^2} \times \hat{\omega}(\xi) \qquad (\xi \neq 0), \qquad \hat{u}(0) = 0,$$

*then* $\operatorname{div} u = 0$ *and* $\operatorname{curl} u = \omega$.

*Proof.* In Fourier space, $\widehat{\operatorname{div} u}(\xi) = i\xi \cdot \hat{u}(\xi) = i\xi \cdot (\xi \times \hat{\omega})/|\xi|^2 = 0$. Likewise

$$\widehat{\operatorname{curl} u}(\xi) = i\xi \times \Big(\frac{i\xi}{|\xi|^2} \times \hat{\omega}\Big) = \hat{\omega} - \frac{\xi(\xi \cdot \hat{\omega})}{|\xi|^2} = \hat{\omega},$$

where the last step uses $\xi \cdot \hat{\omega} = 0$ from $\operatorname{div}\omega = 0$. This identifies $u = K_{\mathrm{BS}} * \omega$ as a Fourier multiplier and proves $\operatorname{div} u = 0$ without a principal-value interchange on $\mathbb{R}^3$. The $\mathbb{R}^3$ interchange remains a separate analytic remainder if the periodisation is not used. □

## 4 Obstruction lemmas: expanded to the missing piece

The lemmas in this section are the hypotheses of Theorem 2.1 that are not classical theorems at the endpoint required by the criterion. Each proof is written until the estimate that is missing.

**Lemma 4.1** (Strain endpoint)**.** *There exist a Banach space $X(\mathbb{T}^3)$ and a constant $C_*$ such that*

$$\|S\|_{L^\infty} \;\le\; C_*\|\omega\|_X$$

*for every smooth divergence-free $\omega$, with $S$ the deformation tensor of $u = K_{\mathrm{BS}} * \omega$.*

*Proof.* The kernel representation is

$$S(x) = \mathrm{P.V.} \int_{\mathbb{T}^3} K(x-y)\omega(y)\, d\lambda(y)$$

with $K$ as in the coefficient-correction section, angular factor

$$A(\theta, \omega) = (\theta \times \omega) \otimes \theta + \theta \otimes (\theta \times \omega).$$

A pointwise bound

$$\|A(\theta,\omega)\|_{\mathrm{op}} \le 2|\theta \times \omega| \le 2|\omega|$$

together with $\operatorname{area}(S^2) = 4\pi$ produces the scale tag $C_{\mathrm{CZ}}(3) = 3/2$. *The following piece is missing to meet Clay standards.* The Calderón–Zygmund operator $\omega \mapsto S$ is bounded on $L^p(\mathbb{T}^3)$ for $1 < p < \infty$, and is not bounded on $L^\infty$. The estimate $\|S\|_{L^\infty} \le \frac{3}{2}\|\omega\|_{L^\infty}$ is therefore false as a general statement. The missing piece is an endpoint bound in a space that does control $\|S\|_{L^\infty}$, classically $B^0_{\infty,1}$ or a Kozono–Taniuchi logarithmic substitute

$$\|S\|_{L^\infty} \le C\|\omega\|_{BMO}\big(1 + \log(1 + \|\omega\|_{H^s})\big).$$

Either substitute changes the Young bookkeeping: the factor $6MI_4$ need not be the stretching contribution. Until that endpoint is fixed and the constant $C_*$ is tracked into $C_{\mathrm{abs}}$, Lemma 4.2 cannot feed Lemma 3.6. Chat fragments that set $C_* = 3/2$ on $L^\infty$ are not used. □

**Lemma 4.2** (Stretching identification)**.** *On a classical solution of* (2)*, after mollification,*

$$\int_{\mathbb{T}^3} \langle \omega_\varepsilon, (\omega_\varepsilon \cdot \nabla) u_\varepsilon \rangle \varphi_\varepsilon \, d\lambda \;\le\; C_* M_\varepsilon I_4(\varphi_\varepsilon) + \mathscr{C}_\varepsilon,$$

*with $\mathscr{C}_\varepsilon \to 0$ as $\varepsilon \to 0$ in the sense used to pass to the limit in $S_\varepsilon$, and with $C_*$ the constant of Lemma 4.1.*

*Proof.* By Lemma 3.1,

$$\langle \omega_\varepsilon, (\omega_\varepsilon \cdot \nabla) u_\varepsilon \rangle = \langle \omega_\varepsilon, S_\varepsilon \omega_\varepsilon \rangle \le \|S_\varepsilon\|_{L^\infty} |\omega_\varepsilon|^2.$$

Integrating against $\varphi_\varepsilon \ge 0$ gives

$$\int \langle \omega_\varepsilon, (\omega_\varepsilon \cdot \nabla) u_\varepsilon \rangle \varphi_\varepsilon \le \|S_\varepsilon\|_{L^\infty} I_4(\varphi_\varepsilon).$$

If Lemma 4.1 supplied $\|S_\varepsilon\|_{L^\infty} \le C_* M_\varepsilon$, the claim would follow with a commutator comparing $S_\varepsilon$ to the strain of $u_\varepsilon = K_{\mathrm{BS}} * \omega_\varepsilon$. *The following piece is missing to meet Clay standards.* No such $L^\infty$ bound is available from the scale tag $3/2$. The missing piece is therefore identical to the missing piece of Lemma 4.1, plus mollifier commutators

$$S(\omega_\varepsilon) - S_\varepsilon(\omega), \qquad (\omega_\varepsilon \cdot \nabla) u_\varepsilon - \eta_\varepsilon * \big((\omega \cdot \nabla) u\big).$$

Those commutators can be written in standard paraproduct form. They vanish as $\varepsilon \to 0$ on a *fixed* classical interval $[0, T] \subset [0, T_*)$, which is legitimate, but the constants obtained that way may depend on $\|u\|_{C^2([0,T]\times\mathbb{T}^3)}$ and therefore cannot be fed into a majorant whose coefficients are required to depend only on $u_0$ and $\nu$. That dependence is the circularity Lemma 4.5 forbids. □

**Lemma 4.3** (Transported weight)**.** *There exists a family of nonnegative scalars $\varphi_\varepsilon$ such that $\|\varphi_\varepsilon(t)\|_{L^\infty}$ is dominated by a function of $t$ whose ODE coefficients depend only on $\|u_0\|_{H^s}$ and $\nu$.*

*Proof.* If $\varphi$ is defined by linear transport

$$\partial_t \varphi + u \cdot \nabla \varphi = 0, \qquad \varphi(\,\cdot\,, 0) = \varphi_0,$$

then $\|\varphi(t)\|_{L^\infty} = \|\varphi_0\|_{L^\infty}$ by the maximum principle for a divergence-free flow. That bound is solution-independent in the weight, and is filled. *The following piece is missing to meet Clay standards.* If the manuscript uses a weight that grows in response to stretching (a quadratic or quartic factor forced by the tether), then $\|\varphi(t)\|_{L^\infty}$ is no longer conserved, and its ODE inherits the same unknown strain constant $C_*$. The missing piece is a source term for $\varphi$ whose size is controlled by initial data alone. Chat fragments that take $\varphi \equiv 1$ make the tether ornamental and move the problem entirely onto Lemma 4.2. Either choice must be declared; they cannot be mixed. □

**Lemma 4.4** (Absorption constant)**.** *The constant $C_{\mathrm{abs}}$ in the schematic inequality*

$$\frac{d}{dt} S_\varepsilon \le C_{\mathrm{abs}} \big(1 + M_\varepsilon^3 \|\varphi_\varepsilon\|_{L^\infty}^{3/2}\big) - \kappa' \int |\omega_\varepsilon|^6 \varphi_\varepsilon \, d\lambda + \mathscr{R}_\varepsilon$$

*depends only on $\kappa = 3/2$, $C_{\mathrm{Sob}}$, $\lambda(\mathbb{T}^3)$, and $\nu$.*

*Proof.* Given Lemmas 3.5, 3.6, 3.7, and a stretching contribution of size $C_* M I_4(\varphi)$, the Young pair produces an explicit $C_{\mathrm{abs}} = C_{\mathrm{abs}}(C_*, C_{\mathrm{Sob}}, \lambda(\mathbb{T}^3))$. With $\kappa = 3/2$ one may compute the algebraic part of that expression. *The following piece is missing to meet Clay standards.* The number $C_*$ is not known to equal $3/2$, and may depend on the solution if a logarithmic endpoint is used. Until Lemma 4.1 supplies a solution-independent $C_*$, $C_{\mathrm{abs}}$ is not a function of $u_0$ and $\nu$ alone. □

**Lemma 4.5** (Non-circular majorant)**.** *There exists a scalar $Y \in C^1([0,\infty);[0,\infty))$ whose ODE*

$$Y' = F(Y), \qquad Y(0) = Y_0(\|u_0\|_{H^s}, \nu)$$

*has coefficients independent of $\{M(t) : t \geq 0\}$, such that $M(t) \leq Y(t)$ on the existence interval.*

*Proof.* Comparison for scalar ODEs is classical: if $M' \leq F(M)$ and $Y' = F(Y)$ with $M(0) \leq Y(0)$, then $M \leq Y$ on the common interval of existence. A clipped cubic $F(y) = C_{\text{abs}}y^2 - \kappa'' y^3$ with $\kappa'' > 0$ even admits global solutions by barrier trapping in $[0, \max(Y_0, C_{\text{abs}}/\kappa'')]$. That comparison step is filled. *The following piece is missing to meet Clay standards.* The differential inequality $M' \leq F(M)$ with an $F$ built only from $u_0$ and $\nu$ is precisely a uniform bound on vortex stretching for arbitrary smooth data. That estimate is the analytic core of the Clay problem. No chat fragment under this account supplies it without either inserting $\sup M$ into $C_{\text{abs}}$ (circular) or modifying the PDE (inadmissible). The missing piece is therefore not a routine ODE lemma. It is the missing a priori bound. □

**Lemma 4.6** (Mollifier limit)**.** *$S_\varepsilon(t) \to S(t)$ and the commutators $\mathscr{C}_\varepsilon, \mathscr{R}_\varepsilon$ vanish as $\varepsilon \to 0$, with constants compatible with Lemma 4.5.*

*Proof.* On any compact classical interval $[0,T] \subset [0,T_*)$, standard mollification yields

$$\omega_\varepsilon \to \omega \quad \text{in } C^k, \qquad S_\varepsilon \to S \quad \text{in } C^{k-1},$$

and commutators vanish. That passage is filled *on a pre-existing classical interval. The following piece is missing to meet Clay standards.* Using those commutator bounds to continue past a putative singularity requires constants that remain controlled as $T \uparrow T_*$. If the commutator estimates depend on $\|u\|_{C^2([0,T]\times\mathbb{T}^3)}$, they explode with $M(t)$ and cannot close Lemma 4.5. The missing piece is a commutator estimate whose constants depend only on $u_0$ and $\nu$. □

**Lemma 4.7** (Jacobi / cocycle of a tethered bracket)**.** *If a bilinear correction $B$ is part of the argument, then $d_2 B = 0$ on the reduced coadjoint orbit.*

*Proof.* The algebraic expansion of a bilinear coadjoint correction into a nine-term cyclic sum is a finite computation in first-order derivatives. Individual pairings can be integrated by parts on $\mathbb{T}^3$ using $\operatorname{div} u = 0$. *The following piece is missing to meet Clay standards.* The missing piece is the verification that the particular kernel proposed for $B$ cancels in every slot while remaining degenerate on $H$ (Lemma 4.8) and while contributing a nonzero negative term to $\frac{d}{dt}M$ (Lemma 4.2). Those three demands are in tension. Chat fragments that treat degeneracy and feedback as simultaneous properties of the same term, without an explicit splitting into a Lyapunov weight versus a modified evolution, are not used. If $B$ is dropped and $\varphi$ is an ordinary weight, this lemma is vacated and Theorem 2.1 is read as a Lyapunov criterion on unmodified Navier–Stokes. □

**Lemma 4.8** (Degeneracy versus feedback)**.** *A correction used as dynamics satisfies $B(F,H) = 0$. A weight used as a Lyapunov device is not required to be a Poisson correction. The two readings are not mixed.*

*Proof.* If $B(\,\cdot\,, H) = 0$, the Hamiltonian evolution is exactly Euler and no extra term appears in (2). Feedback into $\frac{d}{dt}M$ must then come from a Lyapunov functional, not from a modified bracket. If $B(\,\cdot\,, H) \neq 0$, the evolution is not (2) and Theorem 2.1 does not apply to Navier–Stokes. This is a logical dichotomy, not an estimate. It is filled as logic and left open as geometry. □

## 5 What chat fragments contribute, and what they do not

The following fragments from prior conversations are used.

- The ruling that a metriplectic extra damping term changes the PDE and is excluded. That is incorporated as the exact system (1)–(2).
- The ruling that $\kappa = 1$ is illegitimate in three dimensions. That is Lemma 3.4.
- The ruling that an a-posteriori Lyapunov functional on the exact equations is a criterion, not a completed Clay proof. That is the official status of Theorem 2.1.
- Route A: geometric claims, if any, are separated from the analytic criterion. That is Lemma 4.8.

The following fragments are not used.

- Any construction in which $B$ is simultaneously degenerate on $H$ and the source of a new negative term in the vorticity equation.
- Any estimate that inserts $\sup_{t<T} M(t)$ into $C_{\mathrm{abs}}$ or into the existence time of the majorant.
- Any identification of $C_*$ with $3/2$ on $L^\infty$.
- Any claim that filling classical lemmas (transport, viscosity, Kato, BKM, Young) constitutes global regularity.

## 6 What would convert Theorem 2.1 into the Clay theorem

Exactly one additional ingredient is required: a proof of Lemmas 4.1, 4.2, 4.4, 4.5, and 4.6 with constants depending only on $\|u_0\|_{H^s}$ and $\nu$, for arbitrary smooth divergence-free $u_0$. Those five lemmas are the same missing a priori bound under different names. Proving them would prove Theorem 2.1 unconditionally, which is the Clay statement, which is global regularity.

No rearrangement of those names, and no observation that incompressible flow and global regularity share an "enigma," discharges the estimate.

## 7 Summary of filled versus open

| **Lemma** | **Status** | **Role in Theorem 2.1** |
|---|---|---|
| Stretching $= \omega \cdot S\omega$ | filled | algebra |
| Transport IBP on $\mathbb{T}^3$ | filled | cancels convection |
| Viscous IBP | filled | sign of dissipation |
| $\kappa = 3/2$ arithmetic | filled | coefficient lock |
| Hölder on $\mathbb{T}^3$ | filled | gate |
| Young pair | filled algebraically | pending $C_*$ |
| Sobolev $H^1 \hookrightarrow L^6$ | classical, filled | gate |
| Kato local existence | classical, cited | starts the clock |
| BKM continuation | classical, cited | upgrades $\int M < \infty$ |
| $\operatorname{curl} \nabla p = 0$ | filled | pressure drops from vorticity |
| Fourier Biot–Savart on $\mathbb{T}^3$ | filled | $\operatorname{div} u = 0$ |
| Strain endpoint in $L^\infty$ | obstruction | feeds $C_*$ |
| Stretching identification | obstruction | feeds $6MI_4$ |
| Weight independent of $M$ | partial / obstruction | feeds $\varphi$ |
| $C_{\mathrm{abs}}(u_0, \nu)$ only | obstruction | feeds the ODE |
| Non-circular majorant | obstruction | *the* missing bound |
| Mollifier constants vs. $u_0$ | obstruction | passage to the limit |
| Jacobi of $B$ | vacated or obstruction | geometry, not required if $B$ is dropped |